

Soc internship day 31

1. What is Threat Hunting?

Threat hunting is a proactive cybersecurity activity where security analysts actively search for hidden threats or malicious activity that may have bypassed traditional security tools such as antivirus, firewalls, or intrusion detection systems.

Unlike automated detection, threat hunting starts with a hypothesis and investigates systems to identify attackers before they cause significant damage.

Goals of Threat Hunting

- Detect attackers that have evaded security controls.
- Reduce the time attackers remain undetected (dwell time).
- Identify indicators of compromise (IOCs) and attacker behaviors.
- Improve an organization's overall security posture.

Threat Hunting Process

1. **Create a hypothesis**
 - Example: "An attacker may have established persistence on employee laptops."
 2. **Collect data**
 - Endpoint logs
 - Authentication logs
 - Network traffic
 - Process execution logs
 - File system changes
 3. **Investigate**
 - Search for unusual behavior.
 - Compare activity against known attacker techniques.
 4. **Analyze findings**
 - Confirm whether activity is malicious or benign.
 5. **Respond**
 - Remove the threat.
 - Patch vulnerabilities.
 - Improve detection rules.
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2. Attacker Persistence Scenario Analysis

Scenario

A company notices that a compromised user's password was changed and the malware was removed. However, a few days later, suspicious activity resumes from the same computer.

Analysis

This suggests the attacker has established **persistence**, meaning they created a way to regain access even after the initial infection appeared to be removed.

Possible Persistence Techniques

- Registry Run keys
- Scheduled Tasks
- Startup folder entries
- Malicious services
- WMI Event Subscriptions
- New local administrator accounts
- Browser extensions
- DLL hijacking
- SSH keys (Linux)
- Cron jobs (Linux)

Indicators to Look For

- Unknown scheduled tasks
- New startup applications
- Recently created administrator accounts
- Unexpected services running
- Registry modifications
- Repeated outbound connections to the same IP address
- PowerShell or command prompt activity
- Security logs showing repeated logins

MITRE ATT&CK Mapping

Common persistence techniques include:

- **T1053** – Scheduled Task/Job
 - **T1547** – Boot or Logon Autostart Execution
 - **T1136** – Create Account
 - **T1546** – Event Triggered Execution
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3. Threat Hunter Investigation Steps

A threat hunter would:

1. Collect endpoint logs.
2. Review startup programs and services.
3. Check scheduled tasks.
4. Analyze registry changes.
5. Review authentication logs.
6. Examine PowerShell history.

7. Look for unusual network connections.
 8. Search for similar indicators on other systems.
 9. Validate findings with endpoint detection tools.
 10. Remove persistence mechanisms and monitor for recurrence.
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4. Recommended Response

If persistence is confirmed:

- Isolate the affected system.
- Remove all persistence mechanisms.
- Reset passwords for affected accounts.
- Revoke active sessions and tokens.
- Scan the network for lateral movement.
- Patch exploited vulnerabilities.
- Reimage the system if complete removal cannot be guaranteed.
- Increase monitoring and update detection rules.